

Physics Simulation and STRidge Identification Report

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.

1. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	1.00000000e+03	kg m ⁻³
Dynamic viscosity	ν	1.00000000e+01	Pa s
Kinematic viscosity	ν	1.00000000e-02	m ² s ⁻¹
Characteristic velocity	$u_{\max}$	1.00000000e+00	m s ⁻¹
Characteristic length	D	1.00000000e+00	m
Reynolds number	Re	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s ⁻²
Grid points in x	N_x	3.10000000e+01	cells
Grid points in y	N_y	3.10000000e+01	cells
Grid spacing in x	Δx	3.22580645e-02	m
Grid spacing in y	Δy	3.22580645e-02	m
Time step	Δt	1.00000000e-04	s
Numerical threshold	ϵ	7.00000000e-11	-

2. Numerical Discretisation

Quantity	Symbol	Value	Unit
Parameters listed above are the values used by the selected solver.			

3. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(9851, 31, 30)
Regression field shape (t, x, y)	(8051, 28, 29)
Regression samples	6,537,412
Dictionary terms	5
First saved time index	150

Quantity	Value
Temporal crop at each boundary	1800 layers
x-boundary crop	1 points
y-boundary crop	1 points

4. Governing Equation Verification

Reference equation

$$u_t = -1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_{xx} + 1.00000000e-02*u_{yy} - 1.00000000e-03*p_x$$

Learned equation

$$u_t = -9.47336338e-01*uu_x - 6.00121162e-01*vu_y + 1.26019801e-02*u_{xx} + 8.02691012e-03*u_{yy} - 6.37858397e-04*p_x$$

Term	Reference coefficient	Learned coefficient	Percentage error
uu_x	-1.00000000e+00	-9.47336338e-01	-5.26636622e+00%
vu_y	-1.00000000e+00	-6.00121162e-01	-3.99878838e+01%
u_xx	+1.00000000e-02	+1.26019801e-02	2.60198011e+01%
u_yy	+1.00000000e-02	+8.02691012e-03	1.97308988e+01%
p_x	-1.00000000e-03	-6.37858397e-04	-3.62141603e+01%

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	5.87801607e-04
L0 penalty	1.00000000e-06
Training RMSE	9.20238406e-02
Validation RMSE	9.59731897e-02

6. Reproducibility Notes

- Input directory:
/Users/minnie/Desktop/PhysicsFYP/ns_solver/stridge_model/output_4movingwalls/v_velocity_field
- Fields were loaded starting at saved time index 150.
- Derivatives were calculated using second-order edge-aware finite differences via `numpy.gradient`.

- The learned equation is compared term-by-term with the supplied reference coefficients.

7. Transverse Momentum Equation

Reference equation

$$v_t = -1.00000000e+00*uv_x - 1.00000000e+00*vv_y + 1.00000000e-02*v_{xx} + 1.00000000e-02*v_{yy} - 1.00000000e-03*p_y$$

Learned equation

$$v_t = -8.50541042e-01*uv_x - 1.23376054e+00*vv_y + 8.22045496e-03*v_{xx} + 8.06635930e-03*v_{yy} - 1.03326353e-03*p_y$$

Term	Reference coefficient	Learned coefficient	Percentage error
uv_x	-1.00000000e+00	-8.50541042e-01	-1.49458958e+01%
vv_y	-1.00000000e+00	-1.23376054e+00	-2.33760536e+01%
v_xx	+1.00000000e-02	+8.22045496e-03	1.77954504e+01%
v_yy	+1.00000000e-02	+8.06635930e-03	1.93364070e+01%
p_y	-1.00000000e-03	-1.03326353e-03	-3.32635306e+00%

V STRidge Selection and Error Metrics

Metric	Value
Selected tolerance	9.42668455e-04
Training RMSE	8.08743204e-02
Validation RMSE	8.15680328e-02